

Ben Armstrong

EMAIL: research@benarmstrong.ca
WEBSITE: <https://www.benarmstrong.ca>

APRIL 2025 - ONGOING Postdoctoral Researcher, **Tulane University**
Supervisor: Dr. Nicholas Mattei

EDUCATION

- FEBRUARY 2025 PhD in Computer Science, **University of Waterloo**
Thesis: "The Power of Experimental Approaches to Social Choice"
GPA: 91/100 | Supervisor: Dr. Kate Larson
- OCTOBER 2018 Master of Mathematics in COMPUTER SCIENCE, **University of Waterloo**
Thesis: "Coordination in a Peer Production Platform"
GPA: 92/100 | Supervisor: Dr. Kate Larson
- MAY 2016 BSc Honours in COMPUTER SCIENCE, **University of Northern BC**
Thesis: "Software Tools for Agent-Based Evolutionary Modeling"
GPA: 4.02/4.33 during final two years | Advisor: Dr. Jernej Polajnar

PUBLICATIONS

Conferences

- Ratip Emin Berker, Ben Armstrong, Vincent Conitzer, and Nihar B. Shah. "Designing Rules to Pick a Rule: Aggregation by Consistency." The Fourteenth International Conference on Learning Representations (ICLR) 2026. (Forthcoming)
- Joshua Caiata, Ben Armstrong, and Kate Larson. "What Voting Rules Actually Do: A Data-Driven Analysis of Multi-Winner Voting." Proceedings of the 40th Annual AAAI Conference on Artificial Intelligence. 2026.
- Nicole Smith-Vaniz, Harper Lyon, Lorraine Steigner, Ben Armstrong, and Nicholas Mattei. "Investigating Political and Demographic Associations in Large Language Models Through Moral Foundations Theory." Proceedings of the 8th conference on Artificial Intelligence, Ethics, and Society. 2025.
- Ben Armstrong, Shiri Alouf-Heffetz, and Nimrod Talmon. "Optimizing Viscous Democracy." Proceedings of the 33rd International Joint Conference on Artificial Intelligence. 2024.
- Ben Armstrong, and Kate Larson. "Liquid Democracy for Low-Cost Ensemble Pruning." Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems (Extended Abstract). 2024.
- Carter Blair, Ben Armstrong, and Kate Larson. "Liquid Ensemble Selection for Continual Learning." 37th Canadian Conference on Artificial Intelligence. 2024.
- Shiri Alouf-Heffetz, Ben Armstrong, Kate Larson, and Nimrod Talmon. "How Should We Vote? A Comparison of Voting Systems within Social Networks." Proceedings of the 31st International Joint Conference on Artificial Intelligence. 2022.

Working Papers

Leonardo Matone, Ben Abramowitz, Ben Armstrong, Avinash Balakrishnan, Nicholas Mattei. “Deepvoting: Learning Voting Rules with Tailored Embeddings”. arXiv:2408.13630 (2025).

Carter Blair, Ben Armstrong, Shiri Alouf-Heffetz, Nimrod Talmon, and Davide Grossi. “Probably Approximately Consensus: On the Learning Theory of Finding Common Ground” Presented at 2nd Workshop on Social Choice and Learning Algorithms (SCaLA) at 34th International Joint Conference on Artificial Intelligence. 2025.

Workshop and Non-Archival

Amanda Aird, Ben Armstrong, Nicholas Mattei, and Robin Burke. “Envy-Free but Still Unfair: Envy-Freeness Up To One Item (EF-1) in Personalized Recommendation.” arXiv:2509.09037. Presented at the FAccTRec Workshop at 19th ACM Conference on Recommender Systems (RecSys). 2025.

Ben Armstrong, and Kate Larson “On the Limited Applicability of Liquid Democracy”. 3rd Games, Agents, and Incentives Workshop (GAIW) at 20th International Conference on Autonomous Agents and Multiagent Systems. 2021.

Ben Armstrong. “Exploring the Relationship Between Social Choice and Machine Learning.” Doctoral Consortium at the 20th International Conference on Autonomous Agents and Multiagent Systems. 2021.

Ben Armstrong, Kate Larson, “Machine Learning to Strengthen Democracy.” AI For Social Good Workshop at the Neural Information Processing Systems Conference. 2019.

Ben Armstrong, Megan Beretta, Evan Crothers, Michael Karlin, Dongwoo Kim, Justin Longo, Lorne Powell, and Trooper Sanders. “Siri Humphrey: Design Principles for an AI Policy Analyst.” 2019 Summer Institute on AI and Society (published online). 2019.

Edward Parson, Robert Lempert, Ben Armstrong, Evan Crothers, Chad DeChant, and Nick Novelli. “Could AI Drive Transformative Social Progress? What Would This Require?” 2019 Summer Institute on AI and Society (published online) 2019.

Ben Armstrong, Kate Larson, “Approval in the Echo Chamber.” 4th Workshop on Exploring Beyond the Worst Case in Computational Social Choice at 16th International Conference on Autonomous Agents and Multiagent Systems. 2017.

SOFTWARE DEVELOPMENT

LIBRARY	(<i>Under Development</i>) Optimal Voting Python library to learn novel and interpretable voting rules optimized for user-specified learning targets. Has been used to find rules with strong axiomatic properties and generate rankings which minimize disagreement among voters . View at: pypi.org/project/optimal-voting/
LANGUAGES	Experience with Python, Java, C++, C, SQL, Prolog (my first language!), Lisp
FRAMEWORKS	Familiar with Pytorch, Tensorflow, Numpy, Pandas, Pyplot, Optimal Voting

TEACHING

Certifications

UNIVERSITY MATHEMATICS TEACHING TECHNIQUES	A semester-long course aimed at upper level graduate students and new instructors. Covered many fundamental aspects of teaching, modern pedagogical styles, and delivery of lectures to undergraduate students.
FUNDAMENTALS OF UNIVERSITY TEACHING	A University of Waterloo program aimed at graduate students involving several seminars on a range of pedagogical topics and peer-to-peer practice lectures.

Teaching Assistant I have been a teaching assistant, often doing significant portions of the course delivery, for the following courses (across Fall, Winter, Spring terms):

GRADUATE SEMINAR <i>F23, F22, F21, F20, W20</i>	Ethics Seminar for Master's of Data Science and AI Deliver one third of ethics seminar to graduate students studying data science and AI. This involves preparing material from on a syllabus, delivering it to students, and moderating in-class discussion on the content.
CS 492 <i>S23, W23, S22, S21, W21, S19, S18, W18</i>	Social Implications of Computing Moderate in-class debates over various social issues within computing, assist students with essay-writing, mark essays, and track participation.
CS 497 <i>W22</i>	Discrimination in Computing Assist in running class discussions about ethical issues within modern computer science, mark essays, and track participation.
CS 486 <i>F22, F21, F20, W20</i>	Introduction to Artificial Intelligence Mark assignments and aid students within the university's main undergraduate AI course.
SEPT 2015–APR 2016	Teaching Assistant, University of Northern BC Run lab and supervise student projects for CPSC 475 (Multi-Agent Systems). Mark assignments and midterms for Math 220 (Linear Algebra) and Math 320 (Abstract Algebra).

ACTIVITIES

2024, 2025	Workshop on Social Choice and Learning Algorithms Along with co-organizers, I developed the workshop proposal, organized reviewers, and will run the workshop program.
2024	Workflow Chair at 33rd International Joint Conference on Artificial Intelligence (IJCAI) Support reviewers, meta-reviewers, and authors during the review process with all types of questions.
JAN 2018–DEC 2019	AI and Human Rights Advisory Board, Balsillie School I have served as a technical consultant to students developing methods of using artificial intelligence to further human rights and proposals to guide Canada's policies regarding the use of AI.

PROGRAM COMMITTEE

AAAI Conference on Artificial Intelligence (AAAI)

2023, 2024, 2025, 2026

Conference on Neural Information Processing Systems (NeurIPS)

2022, 2023, 2024, 2025

International Conference on Machine Learning (ICML)

2022, 2023, 2024, 2025

International Joint Conference on Artificial Intelligence (IJCAI)

2025

International Conference on Learning Representations (ICLR)

2026

International Conference on Autonomous Agents and Multiagent Systems (AAMAS)

2026

International Conference on Algorithmic Decision Theory (ADT)

2024

European Conference on Artificial Intelligence (ECAI)

2025

Games, Agents, Incentives Workshop (GAIW) at International Conference on Autonomous Agents and Multiagent Systems

2021, 2022, 2023, 2024, 2025

INDUSTRY WORK EXPERIENCE

JAN 2015–AUG 2016

Software Developer, Goldstream Publishing

Developed iOS application to crowd-source tracking of moose populations in British Columbia. Developed iOS application to serve as online hub for the fishing community.

SEPT 2014–DEC 2014

Software Developer, *Democracy Link*

Developed mobile application to provide information on how, where, and when to vote in the 2015 Canadian federal election.

JAN 2014–MAY 2015

Research Assistant, Business Intelligence Research Group at UNBC

Developed and analyzed performance of routing protocols in a real-time distributed database system. Developed business intelligence software for use by health care practitioners.

AWARDS

SEPT 2020–AUG 2023	NSERC Postgraduate Scholarship – Doctoral program Value: \$21,000 / year
SEPT 2019–AUG 2023	President's Graduate Scholarship Value: \$5,000 / year
SEPT 2019	Teaching Assistant Excellence Award Value: \$500
SEPT 2019–AUG 2020	Queen Elizabeth II Graduate Scholarship in Science and Technology Value: \$15,000
SEPT 2018–AUG 2020	David R. Cheriton Graduate Scholarship Value: \$10,000 / year

OTHER

APRIL 2022 – APRIL 2025	Instructor at University of Waterloo Karate & Jujitsu Club I hold a second degree black belt with Seiwa Kai International. I taught one to two classes a week, monitored student's overall progression, and developed short and long term teaching priorities and class delivery strategies.
-------------------------	---